

SAFETY DATA SHEET
according to Regulation (EC) No. 1907/2006,
as amended

CLARIANT 

EMULSOGEN MP2

Page 1(27)

Substance key: 000000287838

Revision Date: 19.07.2024

Date of last issue: 27.09.2021

Version : 1 - 5 / EU

Date of printing : 05.04.2026

SECTION 1: Identification of the substance/mixture and of the company/undertaking

1.1. Product identifier

Trade name

EMULSOGEN MP2

Material number: 185323

Chemical nature:

Mixture of anionic and non ionic emulsifier

1.2. Relevant identified uses of the substance or mixture and uses advised against

Relevant identified uses of the substance or mixture

Industry sector :

Crop Protection

Type of use:

Emulsifier

1.3. Details of the supplier of the safety data sheet

Identification of the company

Clariant Produkte (Deutschland) GmbH

65926 Frankfurt am Main

Telephone no. : +49 69 305 18000

Information about the substance/mixture

BU Care Chemicals

Product Stewardship

e-mail: SDS.Europe@clariant.com

1.4. Emergency telephone number

00800-5121 5121 (24 h)

SECTION 2: Hazards identification

2.1 Classification of the substance or mixture

Classification (REGULATION (EC) No 1272/2008)

Flammable liquids, Category 3

H226: Flammable liquid and vapour.

Skin irritation, Category 2

H315: Causes skin irritation.

Serious eye damage, Category 1

H318: Causes serious eye damage.

Long-term (chronic) aquatic hazard,
Category 3

H412: Harmful to aquatic life with long lasting
effects.

2.2 Label elements

Labelling (REGULATION (EC) No 1272/2008)

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Hazard pictograms :



Signal word :

Danger

Hazard statements :

H226 Flammable liquid and vapour.
H315 Causes skin irritation.
H318 Causes serious eye damage.
H412 Harmful to aquatic life with long lasting effects.

Precautionary statements :

Prevention:

P210 Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking.
P273 Avoid release to the environment.
P280 Wear protective gloves/ eye protection/ face protection.

Response:

P303 + P361 + P353 IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water/ shower.
P305 + P351 + P338 + P310 IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing. Immediately call a POISON CENTER/ doctor.
P370 + P378 In case of fire: Use dry sand, dry chemical or alcohol-resistant foam to extinguish.

Disposal:

P501 Dispose of contents/ container to an approved waste disposal plant.

Hazardous components which must be listed on the label:

Benzenesulfonic acid, mono-C11-13-branched alkyl derivs., calcium salts

2.3 Other hazards

This substance/mixture contains no components considered to be either persistent, bioaccumulative and toxic (PBT), or very persistent and very bioaccumulative (vPvB) at levels of 0.1% or higher.

Ecological information: The substance/mixture does not contain components considered to have endocrine disrupting properties according to REACH Article 57(f) or Commission Delegated regulation (EU) 2017/2100 or Commission Regulation (EU) 2018/605 at levels of 0.1% or higher.

Toxicological information: The substance/mixture does not contain components considered to have endocrine disrupting properties according to REACH Article 57(f) or Commission Delegated regulation (EU) 2017/2100 or Commission Regulation (EU) 2018/605 at levels of 0.1% or higher.

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SECTION 3: Composition/information on ingredients

3.2 Mixtures

Components

Chemical name	CAS-No. EC-No. Index-No. Registration number	Classification	Concentration (% w/w)
Benzenesulfonic acid, mono-C11-13-branched alkyl derivs., calcium salts	68953-96-8 273-234-6 01-2119964467-24 01-2119964467-24-0001	Acute Tox. 4; H312 Skin Irrit. 2; H315 Eye Dam. 1; H318 Aquatic Chronic 2; H411	>= 15 - < 20
Iso-Butanol	78-83-1 201-148-0 603-108-00-1 01-2119484609-23 01-2119484609-23-0000 01-2119484609-23-0049	Flam. Liq. 3; H226 Skin Irrit. 2; H315 Eye Dam. 1; H318 STOT SE 3; H336 (Central nervous system) STOT SE 3; H335 (Respiratory system)	>= 5 - < 10

For explanation of abbreviations see section 16.

SECTION 4: First aid measures

4.1 Description of first aid measures

- General advice : Remove/ Take off immediately all contaminated clothing.
Get medical advice/ attention if you feel unwell.
- If inhaled : If inhaled, remove to fresh air.
Get medical advice/ attention.
- In case of skin contact : Wash off immediately with plenty of water for at least 15 minutes.
Consult a physician.
- In case of eye contact : Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes.
Consult a physician.
- If swallowed : If swallowed do not induce vomiting, seek medical advice and show safety datasheet or label

4.2 Most important symptoms and effects, both acute and delayed

- Symptoms : corrosive effects
- Risks : Causes skin irritation.
Causes serious eye damage.

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4.3 Indication of any immediate medical attention and special treatment needed

Treatment : Treat symptomatically.

SECTION 5: Firefighting measures

5.1 Extinguishing media

Suitable extinguishing media : Water spray jet
Alcohol-resistant foam
Carbon dioxide (CO₂)
Dry powder

Unsuitable extinguishing media : High volume water jet

5.2 Special hazards arising from the substance or mixture

Specific hazards during firefighting : In case of fire hazardous decomposition products may be produced such as:
Sulphur dioxide
Carbon monoxide
Carbon dioxide (CO₂)

5.3 Advice for firefighters

Special protective equipment for firefighters : Self-contained breathing apparatus

Further information : Wear suitable protective equipment.

SECTION 6: Accidental release measures

6.1 Personal precautions, protective equipment and emergency procedures

Personal precautions : Ensure adequate ventilation.
Wear suitable protective clothing.
Keep away sources of ignition.

6.2 Environmental precautions

Environmental precautions : Do not empty into drains.

6.3 Methods and material for containment and cleaning up

Methods for cleaning up : Pick up with absorbent material (eg sand, sawdust). Rinse away rest with water
Can be landfilled or incinerated, when in compliance with local regulations.

6.4 Reference to other sections

Information regarding Safe handling, see chapter 7., For personal protection see section 8., For disposal considerations see section 13.

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SECTION 7: Handling and storage

7.1 Precautions for safe handling

- Advice on safe handling : Provide adequate ventilation.
Handle and open container with care.
- Advice on protection against fire and explosion : Keep away from sources of ignition - No smoking. Take precautionary measures against build-up of electrostatic charges, e.g earthing during loading and off-loading operations.
- Hygiene measures : Wash hands before breaks and at the end of workday. Use protective skin cream before handling the product. Take off immediately all contaminated clothing and wash it before reuse.

7.2 Conditions for safe storage, including any incompatibilities

- Further information on storage conditions : Keep container tightly closed and in a well-ventilated place.
Keep away from heat. Keep away from direct sunlight.

7.3 Specific end use(s)

- Specific use(s) : No further recommendations.

SECTION 8: Exposure controls/personal protection

8.1 Control parameters

Occupational Exposure Limits

Derived No Effect Level (DNEL) according to Regulation (EC) No. 1907/2006:

Substance name	End Use	Exposure routes	Potential health effects	Value
Iso-Butanol CAS-No.: 78-83-1	Workers	Inhalation	Long-term local effects	310 mg/m3
	Remarks:DNEL			
	General population	Inhalation	Long-term local effects	55 mg/m3
	Remarks:DNEL			
Benzenesulfonic acid, mono-C11-13-branched alkyl derivs., calcium salts CAS-No.: 68953-96-8	Workers	Inhalation	Long-term systemic effects	6 mg/m3
	Remarks:DNEL			
	Workers	Dermal	Long-term systemic effects	0,85 mg/kg bw/day
	Remarks:DNEL			
	Consumers	Inhalation	Long-term systemic	1,48 mg/m3

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			effects	
	Remarks:DNEL			
	Consumers	Dermal	Long-term systemic effects	0,425 mg/kg bw/day
	Remarks:DNEL			
	Consumers	Oral	Long-term systemic effects	0,43 mg/kg bw/day
	Remarks:DNEL			

Predicted No Effect Concentration (PNEC) according to Regulation (EC) No. 1907/2006:

Substance name	Environmental Compartment	Value
Benzenesulfonic acid, mono-C11-13-branched alkyl derivs., calcium salts CAS-No.: 68953-96-8	Fresh water	0,023 mg/l
	Water (intermittent release)	0,29 mg/l
	Marine water	0,002 mg/l
	Sewage treatment plant	5,5 mg/l
	Fresh water sediment	1,35 mg/kg dry weight (d.w.)
	Marine sediment	0,135 mg/kg dry weight (d.w.)
	Soil	0,124 mg/kg dry weight (d.w.)

8.2 Exposure controls

Engineering measures

Use engineering controls such as local or general exhaust to maintain airborne concentrations below exposure limits.

Personal protective equipment

Eye/face protection : Safety glasses

Hand protection

Break through time : 480 min

Glove thickness : 0,7 mm

Remarks : Long-term exposure Impervious butyl rubber gloves

Break through time : 30 min

Glove thickness : 0,4 mm

Remarks : For short-term exposure (splash protection): Nitrile rubber gloves.

Skin and body protection : Wear suitable protective equipment.

Respiratory protection : Use respiratory protection in case of insufficient exhaust ventilation or prolonged exposure

Protective measures : Avoid contact with skin and eyes.
Do not inhale vapours

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SECTION 9: Physical and chemical properties

9.1 Information on basic physical and chemical properties

Physical state	: Liquid
Colour	: yellow-brown
Odour	: of isobutanol
Freezing point	: no data available
Boiling point	: no data available
Flammability	: Not applicable
Upper explosion limit / upper flammability limit	: no data available
Lower explosion limit / Lower flammability limit	: no data available
Flash point	: approx. 55 °C
Auto-ignition temperature	: no data available
Decomposition temperature	: not tested.
pH	: 6 - 8 (28 °C) Concentration: 5 % Ethanol/Water
Viscosity	
Viscosity, dynamic	: < 5.000 mPa.s (25 °C)
Viscosity, kinematic	: no data available
Solubility(ies)	
Water solubility	: (20 °C) soluble

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Partition coefficient: n-
octanol/water : no data available

Vapour pressure : no data available

Density : ca. 0,99 g/cm³ (20 °C)

Relative vapour density : no data available

Particle characteristics
Particle size : Not applicable

9.2 Other information

Metal corrosion rate : Not applicable

SECTION 10: Stability and reactivity

10.1 Reactivity

See section 10.3. "Possibility of hazardous reactions"

10.2 Chemical stability

Stable under normal conditions.

10.3 Possibility of hazardous reactions

Hazardous reactions : Uncleaned empty vessels may contain product gases which can form explosive mixtures with air.

10.4 Conditions to avoid

Conditions to avoid : Keep away from heat and sources of ignition.

10.5 Incompatible materials

Materials to avoid : not known

10.6 Hazardous decomposition products

When handled and stored appropriately, no dangerous decomposition products are known

SECTION 11: Toxicological information

11.1 Information on hazard classes as defined in Regulation (EC) No 1272/2008

Acute toxicity

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Acute oral toxicity : LD50: > 2.000 mg/kg
Method: OECD Test Guideline 401

Acute inhalation toxicity : Remarks: no data available

Acute dermal toxicity : Acute toxicity estimate: > 2.000 mg/kg
Method: Calculation method

Components:

Benzenesulfonic acid, mono-C11-13-branched alkyl derivs., calcium salts:

Acute oral toxicity : LD50 (Rat, male and female): > 2.000 mg/kg
Method: OECD Test Guideline 401
GLP: yes
Assessment: The substance or mixture has no acute oral toxicity

Acute inhalation toxicity : Remarks: not required

Acute dermal toxicity : LD50 (Rat, male and female): 1.000 - 1.600 mg/kg
Method: OECD Test Guideline 402
GLP: yes
Assessment: The component/mixture is moderately toxic after single contact with skin.

Iso-Butanol:

Acute oral toxicity : LD50 (Rat, male and female): 2.830 - 3.350 mg/kg
Method: OECD Test Guideline 401
GLP: yes
Assessment: The substance or mixture has no acute oral toxicity

Acute inhalation toxicity : LC50 (Rat, male and female): > 18,18 mg/l
Exposure time: 6 h
Test atmosphere: vapour
Method: Other
GLP: yes
Assessment: The substance or mixture has no acute inhalation toxicity

Acute dermal toxicity : LD50 (Rabbit, male and female): 2.460 mg/kg
Method: OECD Test Guideline 402
GLP: yes
Assessment: The substance or mixture has no acute dermal toxicity

Skin corrosion/irritation

Product:

Result : Irritating to skin.

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Components:

Benzenesulfonic acid, mono-C11-13-branched alkyl derivs., calcium salts:

Species	: Rabbit
Exposure time	: 4 h
Assessment	: Irritating to skin.
Method	: Other
Result	: Irritating to skin.
GLP	: yes

Iso-Butanol:

Species	: Rabbit
Exposure time	: 4 h
Method	: OECD Test Guideline 404
Result	: Irritating to skin.
GLP	: yes

Serious eye damage/eye irritation

Product:

Result	: Risk of serious damage to eyes.
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Components:

Benzenesulfonic acid, mono-C11-13-branched alkyl derivs., calcium salts:

Species	: Rabbit
Exposure time	: 1 h - 14 d
Assessment	: Risk of serious damage to eyes.
Method	: Other
Result	: Risk of serious damage to eyes.
GLP	: yes

Iso-Butanol:

Species	: Rabbit
Exposure time	: 24 h
Assessment	: Risk of serious damage to eyes.
Method	: OECD Test Guideline 405
Result	: Risk of serious damage to eyes.
GLP	: yes

Respiratory or skin sensitisation

Product:

Species	: Guinea pig
Method	: Magnusson/Kligman
Result	: No sensitisation effects are known

Components:

Benzenesulfonic acid, mono-C11-13-branched alkyl derivs., calcium salts:

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Test Type : Maximisation Test
Exposure routes : Dermal
Species : Guinea pig
Method : OECD Test Guideline 406
Result : Not a skin sensitizer.
GLP : yes

Assessment : Harmful in contact with skin., Causes skin irritation., Causes serious eye damage.

Iso-Butanol:

Exposure routes : Skin contact
Method : QSAR
Result : Not a skin sensitizer.

Assessment : Causes skin irritation., Causes serious eye damage.

Germ cell mutagenicity

Product:

Germ cell mutagenicity-
Assessment : Not mutagenic in Ames Test

Components:

Benzenesulfonic acid, mono-C11-13-branched alkyl derivs., calcium salts:

Genotoxicity in vitro : Test Type: Ames test
Test system: Salmonella typhimurium
Concentration: 4 - 10000 µg/plate
Metabolic activation: with and without metabolic activation
Method: OECD Test Guideline 471
Result: negative
GLP: yes

Test Type: In vitro gene mutation study in mammalian cells
Test system: Chinese hamster ovary cells
Concentration: 0,6 - 60 µg/ml
Metabolic activation: with and without metabolic activation
Method: OECD Test Guideline 476
Result: negative
GLP: yes
Remarks: By analogy with a product of similar composition

Genotoxicity in vivo : Test Type: Micronucleus test
Species: Mouse (male and female)
Strain: NMRI
Application Route: oral (gavage)
Exposure time: 72 h
Dose: 1122 mg/kg
Method: OECD Test Guideline 474

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Result: negative

GLP: yes

Remarks: By analogy with a product of similar composition

Germ cell mutagenicity-
Assessment : In vitro tests did not show mutagenic effects, In vivo tests did not show mutagenic effects

Iso-Butanol:

Genotoxicity in vitro : Test Type: Ames test
Test system: Salmonella typhimurium
Concentration: 100- 10000 µg/plate
Metabolic activation: with and without metabolic activation
Method: OECD Test Guideline 471
Result: negative
GLP: No information available.

Test Type: Chromosome aberration test in vitro
Test system: Chinese hamster lung cells
Concentration: 11 - 53 mM
Metabolic activation: without
Method: Other
Result: negative
GLP: No information available.

Test Type: In vitro gene mutation study in mammalian cells
Test system: Chinese hamster lung cells
Concentration: <= 107 mM
Metabolic activation: with and without metabolic activation
Method: Other
Result: negative
GLP: No information available.

Genotoxicity in vivo : Test Type: Micronucleus test
Species: Mouse (male and female)
Strain: NMRI
Cell type: Bone marrow
Application Route: oral (gavage)
Exposure time: single iapplication
Dose: 500 - 1000 - 2000 mg/kg
Method: OECD Test Guideline 474
Result: negative
GLP: yes

Germ cell mutagenicity-
Assessment : In vitro tests did not show mutagenic effects, In vivo tests did not show mutagenic effects

Carcinogenicity

Components:

Benzenesulfonic acid, mono-C11-13-branched alkyl derivs., calcium salts:

Carcinogenicity - : No information available.

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Assessment

Iso-Butanol:

Carcinogenicity -
Assessment : No information available.

Reproductive toxicity

Components:

Benzenesulfonic acid, mono-C11-13-branched alkyl derivs., calcium salts:

Effects on fertility : Test Type: Three-generation study
Species: Rat, male and female
Strain: Other
Application Route: oral (feed)
Dose: 14 - 70 - 350 mg/kg bw d
Duration of Single Treatment: 720 d
General Toxicity - Parent: NOAEL: 350 mg/kg body weight
General Toxicity F1: NOAEL: 350 mg/kg body weight
General Toxicity F2: NOAEL: 350 mg/kg body weight
Method: Other
GLP: no
Remarks: By analogy with a product of similar composition

Effects on foetal
development : Test Type: Pre-natal
Species: Mouse, female
Strain: CD1
Application Route: oral (gavage)
Dose: 0,2 - 2,0 - 300 - 600 mg/kg
Duration of Single Treatment: 9 d
General Toxicity Maternal: NOAEL: 2 mg/kg body weight
Teratogenicity: NOAEL: 300 mg/kg body weight
Method: Other
GLP: No information available.
Remarks: By analogy with a product of similar composition

Iso-Butanol:

Effects on fertility : Test Type: Two-generation study
Species: Rat, male and female
Strain: Sprague-Dawley
Application Route: Inhalation
Dose: 500 - 1000 - 2500 ppm
Duration of Single Treatment: 6 h
General Toxicity - Parent: NOAEL: $\geq 7,5$ mg/l
General Toxicity F1: NOAEL: $\geq 7,5$ mg/l
General Toxicity F2: NOAEL: $\geq 7,5$ mg/l
Method: OPPTS 870.3800
GLP: yes

Effects on foetal
development : Test Type: Pre-natal
Species: Rat

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Strain: wistar
Application Route: Inhalation
Dose: 0,5 - 2,5 - 10 mg/l
Duration of Single Treatment: 6 h
General Toxicity Maternal: NOAEL: 10 mg/l
Teratogenicity: NOAEL: 10 mg/l
Method: OECD Test Guideline 414
GLP: yes

Reproductive toxicity - Assessment : No reproductive toxicity to be expected.
No teratogenic effects to be expected.

STOT - single exposure

Product:

Remarks : no data available

Components:

Benzenesulfonic acid, mono-C11-13-branched alkyl derivs., calcium salts:

Assessment : The substance or mixture is not classified as specific target organ toxicant, single exposure.

Iso-Butanol:

Exposure routes : Inhalation
Target Organs : Respiratory Tract
Assessment : May cause respiratory irritation.

Exposure routes : Inhalation
Target Organs : Central nervous system
Assessment : May cause drowsiness or dizziness.

STOT - repeated exposure

Product:

Remarks : no data available

Components:

Benzenesulfonic acid, mono-C11-13-branched alkyl derivs., calcium salts:

Assessment : The substance or mixture is not classified as specific target organ toxicant, repeated exposure.

Iso-Butanol:

Assessment : The substance or mixture is not classified as specific target organ toxicant, repeated exposure.

Repeated dose toxicity

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Remarks : no data available

Components:

Benzenesulfonic acid, mono-C11-13-branched alkyl derivs., calcium salts:

Species : Rat, male and female
NOAEL : 50 mg/kg bw/day
LOAEL : 250 mg/kg
Application Route : oral (feed)
Exposure time : 84 d
Number of exposures : daily in feed
Dose : 50 - 250 mg/kg bw d
Control Group : yes
Method : Other
GLP : no
Remarks : By analogy with a product of similar composition

Species : Rat, male and female
NOAEL : 85 mg/kg bw/day
LOAEL : 145 mg/kg
Application Route : Drinking water
Exposure time : 270 d
Number of exposures : daily in trinking water
Dose : 85 - 145 - 430 mg/kg bw d
Control Group : yes
Method : Other
GLP : no
Remarks : By analogy with a product of similar composition

Repeated dose toxicity -
Assessment : Harmful in contact with skin., Causes skin irritation., Causes serious eye damage.

Iso-Butanol:

Species : Rat, male and female
NOAEL : > 1450 mg/kg bw/day
Application Route : Drinking water
Exposure time : 90 d
Number of exposures : continuous
Dose : 1000 - 4000 - 16000 ppm
Control Group : yes
Method : OECD Test Guideline 408
GLP : yes

Species : Rat, male and female
NOAEL : ca. 3 mg/l
Application Route : Inhalation
Exposure time : 102 d
Number of exposures : 6 hours/day, 5 days/week
Dose : 250 - 1000 - 2500 ppm
Control Group : yes
Method : Other
GLP : yes

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Application Route : Skin contact
Remarks : not tested.

Repeated dose toxicity - Assessment : Causes skin irritation., Causes serious eye damage.

Aspiration toxicity

Product:

no data available

Components:

Benzenesulfonic acid, mono-C11-13-branched alkyl derivs., calcium salts:

No aspiration toxicity classification

Iso-Butanol:

No aspiration toxicity classification

11.2 Information on other hazards

Endocrine disrupting properties

Product:

Assessment : The substance/mixture does not contain components considered to have endocrine disrupting properties according to REACH Article 57(f) or Commission Delegated regulation (EU) 2017/2100 or Commission Regulation (EU) 2018/605 at levels of 0.1% or higher.

SECTION 12: Ecological information

12.1 Toxicity

Product:

Toxicity to fish : LC50 (Danio rerio (zebra fish)): 10 - 100 mg/l
Exposure time: 96 h
Method: OECD Test Guideline 203

Toxicity to daphnia and other aquatic invertebrates : EC50 (Daphnia magna (Water flea)): > 1.000 mg/l
Exposure time: 48 h
Method: OECD Test Guideline 202

Toxicity to algae/aquatic plants : Remarks: no data available

Toxicity to microorganisms : EC50 (Pseudomonas putida): > 1.000 mg/l
Method: DIN 38412 T.8

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Components:

Benzenesulfonic acid, mono-C11-13-branched alkyl derivs., calcium salts:

Toxicity to fish : LC50 (Danio rerio (zebra fish)): 31,6 mg/l
End point: mortality
Exposure time: 96 h
Test Type: static test
Analytical monitoring: no
Method: OECD Test Guideline 203
GLP: yes

Toxicity to daphnia and other : EC50 (Daphnia magna (Water flea)): 62 mg/l
aquatic invertebrates
End point: mortality
Exposure time: 48 h
Test Type: static test
Analytical monitoring: yes
Method: OECD Test Guideline 202
GLP: yes

Toxicity to algae/aquatic : ErC50 (Pseudokirchneriella subcapitata (green algae)): 29
plants
mg/l
End point: Growth rate
Exposure time: 96 h
Test Type: static test
Analytical monitoring: no
Method: Other
GLP: no
Remarks: By analogy with a product of similar composition

NOEC (Pseudokirchneriella subcapitata (green algae)): 0,5
mg/l
End point: Growth rate
Exposure time: 96 h
Test Type: static test
Analytical monitoring: no
Method: Other
GLP: no
Remarks: By analogy with a product of similar composition

Toxicity to microorganisms : EC50 (activated sludge): 550 - 760 mg/l
End point: Bacteria toxicity (respiration inhibition)
Exposure time: 3 h
Test Type: aquatic
Analytical monitoring: yes
Method: OECD Test Guideline 209
GLP: no data available

Toxicity to fish (Chronic : NOEC: 0,23 mg/l
toxicity)
End point: Other
Exposure time: 72 d
Species: Oncorhynchus mykiss (rainbow trout)
Test Type: flow-through test
Analytical monitoring: yes

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Method: Other

GLP: yes

Remarks: By analogy with a product of similar composition

Toxicity to soil dwelling
organisms

: Test Type: artificial soil
LC50: > 1.000 mg/kg
Exposure time: 14 d
End point: mortality
Species: Eisenia fetida (earthworms)
Method: OECD Test Guideline 207
GLP:yes
Remarks: By analogy with a product of similar composition

Test Type: artificial soil
NOEC: 250 mg/kg
Exposure time: 14 d
End point: Body weight
Species: Eisenia fetida (earthworms)
Method: OECD Test Guideline 207
GLP:yes
Remarks: By analogy with a product of similar composition

Test Type: artificial soil
EC10: 82 - 100 mg/kg
Exposure time: 21 d
End point: Reproduction
Species: Other
Method: ISO 11267
GLP:no data available
Remarks: By analogy with a product of similar composition

Plant toxicity

: EC50: 167 mg/kg
Exposure time: 21 d
End point: Growth
Species: Sorghum bicolor (sorghum)
Analytical monitoring: no
Method: OECD Guide-line 208
GLP:yes
Remarks: By analogy with a product of similar composition

NOEC: 100 mg/kg
Exposure time: 21 d
End point: Growth
Species: Sorghum bicolor (sorghum)
Analytical monitoring: no
Method: OECD Guide-line 208
GLP:yes
Remarks: By analogy with a product of similar composition

EC50: 316 mg/kg
Exposure time: 21 d
End point: Growth
Species: Phaseolus aureus

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Analytical monitoring: no
Method: OECD Guide-line 208
GLP:yes
Remarks: By analogy with a product of similar composition

NOEC: 100 mg/kg
Exposure time: 21 d
End point: Growth
Species: Phaseolus aureus
Analytical monitoring: no
Method: OECD Guide-line 208
GLP:yes
Remarks: By analogy with a product of similar composition

Sediment toxicity : Remarks: not tested.

Ecotoxicology Assessment

Chronic aquatic toxicity : Toxic to aquatic life with long lasting effects.

Iso-Butanol:

Toxicity to fish : LC50 (Pimephales promelas (fathead minnow)): 1.430 mg/l
End point: mortality
Exposure time: 96 h
Test Type: flow-through test
Analytical monitoring: yes
Method: Other
GLP: no

Toxicity to daphnia and other aquatic invertebrates : EC50 (Daphnia pulex (Water flea)): 1.100 mg/l
End point: Immobilization
Exposure time: 48 h
Test Type: static test
Analytical monitoring: no
Method: Other
GLP: no

Toxicity to algae/aquatic plants : ErC50 (Pseudokirchneriella subcapitata (green algae)): 1.799 mg/l
End point: Growth rate
Exposure time: 72 h
Test Type: static test
Analytical monitoring: yes
Method: OECD Test Guideline 201
GLP: yes

Toxicity to microorganisms : IC50 (Natural microorganism): 670 mg/l
End point: Growth rate
Exposure time: 16 h
Test Type: static test
Analytical monitoring: no
Method: Other

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GLP: No information available.

Toxicity to fish (Chronic toxicity) : Remarks: not required

Toxicity to daphnia and other aquatic invertebrates (Chronic toxicity) : NOEC: 20 mg/l
End point: Reproduction rate
Exposure time: 21 d
Species: Daphnia magna (Water flea)
Analytical monitoring: yes
Method: Other
GLP: no
Remarks: The details of the toxic effect relate to the nominal concentration.

12.2 Persistence and degradability

Product:

Biodegradability : Biodegradation: 30 - 70 %
Method: OECD Test Guideline 302B

Components:

Benzenesulfonic acid, mono-C11-13-branched alkyl derivs., calcium salts:

Biodegradability : Test Type: aerobic
Inoculum: activated sludge
Concentration: 22 mg/l
Result: Not readily biodegradable.
Biodegradation: 2,9 %
Related to: Carbon dioxide (CO₂)
Exposure time: 28 d
Method: OECD Test Guideline 301E
GLP: no

Physico-chemical removability : Remarks: Not readily biodegradable.

Stability in water : Test Type: abiotic
Degradation half life (Aqueous buffer solution.): > 365 d (50 °C)
pH: 4
Hydrolysis: (16,7 %)
Method: OECD Test Guideline 111
GLP: yes

Test Type: abiotic
Degradation half life (Aqueous buffer solution.): > 365 d (50 °C)
pH: 7
Hydrolysis: (25,6 %)
Method: OECD Test Guideline 111
GLP: yes

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Test Type: abiotic
Degradation half life (Aqueous buffer solution.): > 365 d (50 °C)
pH: 9
Hydrolysis: (18,6 %)
Method: OECD Test Guideline 111
GLP: yes

Iso-Butanol:

Biodegradability : Test Type: aerobic
Inoculum: activated sludge
Result: Readily biodegradable.
Biodegradation: 70 - 80 %
Related to: BOD in % of theoretical OD
Exposure time: 28 d
Method: OECD Test Guideline 301D
GLP: no

12.3 Bioaccumulative potential

Product:

Bioaccumulation : Remarks: no data available

Components:

Benzenesulfonic acid, mono-C11-13-branched alkyl derivs., calcium salts:

Bioaccumulation : Species: Other
Bioconcentration factor (BCF): 3,16
Method: Other
GLP: no
Remarks: Does not bioaccumulate.
The value is given based on a SAR/AAR approach using OECD Toolbox, DEREK, VEGA QSAR models (CAESAR models), etc.

Partition coefficient: n-octanol/water : log Pow: 4,595 (20 °C)
Method: Regulation (EC) No. 440/2008, Annex, A.8
GLP: yes

Iso-Butanol:

Bioaccumulation : Remarks: Due to the low logPow bioaccumulation is not expected

Partition coefficient: n-octanol/water : log Pow: 1 (25 °C)
Method: OECD Test Guideline 117
GLP: yes

12.4 Mobility in soil

Product:

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Distribution among environmental compartments : Remarks: no data available

Components:

Benzenesulfonic acid, mono-C11-13-branched alkyl derivs., calcium salts:

Distribution among environmental compartments : adsorption
Medium: Soil
log Koc: > 5
Method: OECD Test Guideline 121
Remarks: Based upon the calculated log Koc, adsorption to the soil phase is not expected.

Iso-Butanol:

Distribution among environmental compartments : Adsorption/Soil
Medium: water - soil
log Koc: 0,47
Method: estimated

12.5 Results of PBT and vPvB assessment

Product:

Assessment : This substance/mixture contains no components considered to be either persistent, bioaccumulative and toxic (PBT), or very persistent and very bioaccumulative (vPvB) at levels of 0.1% or higher.

Components:

Benzenesulfonic acid, mono-C11-13-branched alkyl derivs., calcium salts:

Assessment : The substance is not identified as a PBT or as a vPvB substance.

Iso-Butanol:

Assessment : Substance is not persistent, bioaccumulative, and toxic (PBT).

12.6 Endocrine disrupting properties

Product:

Assessment : The substance/mixture does not contain components considered to have endocrine disrupting properties according to REACH Article 57(f) or Commission Delegated regulation (EU) 2017/2100 or Commission Regulation (EU) 2018/605 at levels of 0.1% or higher.

12.7 Other adverse effects

Product:

Additional ecological information : Ecological data given refer to the main component.

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Components:

Benzenesulfonic acid, mono-C11-13-branched alkyl derivs., calcium salts:

Environmental fate and pathways : not available

Iso-Butanol:

Environmental fate and pathways : not available

Additional ecological information : Do not allow to enter ground water, waterways or waste water.

SECTION 13: Disposal considerations

13.1 Waste treatment methods

Product : Dispose of in accordance with local regulations.
In accordance with local authority regulations, take to special waste incineration or chemical/physical treatment plant

Contaminated packaging : Packaging that cannot be cleaned should be disposed of as product waste

SECTION 14: Transport information

Section 14.1. to 14.5.

ADR

UN no. UN 1993
Proper shipping name: Flammable liquid, n.o.s.
Hazard inducer(s): Isobutanol
Tetrapropylene benzene sulfonate calcium
Class: 3
Primary risk: 3
Packing group: III
Hazard no. : 30
Remarks Shipment permitted

ADN

UN no. UN 1993
Proper shipping name: Flammable liquid, n.o.s.
Hazard inducer(s): Isobutanol
Tetrapropylene benzene sulfonate calcium
Class: 3
Primary risk: 3
Packing group: III
Remarks Shipment permitted

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RID

UN no.	UN 1993
Proper shipping name:	Flammable liquid, n.o.s.
Hazard inducer(s):	Isobutanol Tetrapropylene benzene sulfonate calcium
Class:	3
Primary risk:	3
Packing group:	III
Hazard no. :	30
Remarks	Shipment permitted

IATA

UN no.	UN 1993
Proper shipping name:	Flammable liquid, n.o.s.
Hazard inducer(s):	Isobutanol Tetrapropylene benzene sulfonate calcium
Class:	3
Primary risk:	3
Packing group:	III
Remarks	Shipment permitted

IMDG

UN no.	UN 1993
Proper shipping name:	Flammable liquid, n.o.s.
Hazard inducer(s):	Isobutanol Tetrapropylene benzene sulfonate calcium
Class:	3
Primary risk:	3
Packing group:	III
Remarks	Shipment permitted
EmS :	F-E S-E

14.6. Special precautions for user

See sections 6 to 8 of this Safety Data Sheet.

14.7. Maritime transport in bulk according to IMO instruments

No transport as bulk according IBC - Code.

SECTION 15: Regulatory information

15.1 Safety, health and environmental regulations/legislation specific for the substance or mixture

REACH - Restrictions on the manufacture, placing on the market and use of certain dangerous substances, mixtures and articles (Annex XVII)

: Conditions of restriction for the following entries should be considered:
Number on list 75, 3

If you intend to use this product as tattoo ink, please contact your vendor.

REACH - Candidate List of Substances of Very High

: Not applicable

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Concern for Authorisation (Article 59).

Regulation (EC) on substances that deplete the ozone layer : Not applicable

Regulation (EU) 2019/1021 on persistent organic pollutants (recast) : Not applicable

Council Regulation (EC) No 111/2005 laying down rules for the monitoring of trade between the Community and third countries in drug precursors : Neither banned nor restricted

Council Regulation (EC) No 273/2004 on drug precursors : Not applicable

Regulation (EU) No 649/2012 of the European Parliament and the Council concerning the export and import of dangerous chemicals : Not applicable

REACH - List of substances subject to authorisation (Annex XIV) : Not applicable

Seveso III: Directive 2012/18/EU of the European Parliament and of the Council on the control of major-accident hazards involving dangerous substances. P5c FLAMMABLE LIQUIDS

Other regulations:

Apart from the data/regulations specified in this chapter, no further information is available concerning safety, health and environmental protection.

Take note of Dir 94/33/EC on the protection of young people at work.
Occupational restrictions for pregnant and breast feeding women

15.2 Chemical safety assessment

No Chemical Safety Assessment (CSA) is yet available for the substance, or for the component substances, contained in this product.

SECTION 16: Other information

Full text of H-Statements

H226 : Flammable liquid and vapour.
H312 : Harmful in contact with skin.
H315 : Causes skin irritation.
H318 : Causes serious eye damage.
H335 : May cause respiratory irritation.
H336 : May cause drowsiness or dizziness.
H411 : Toxic to aquatic life with long lasting effects.

Full text of other abbreviations

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Acute Tox.	: Acute toxicity
Aquatic Chronic	: Long-term (chronic) aquatic hazard
Eye Dam.	: Serious eye damage
Flam. Liq.	: Flammable liquids
Skin Irrit.	: Skin irritation
STOT SE	: Specific target organ toxicity - single exposure

ADN - European Agreement concerning the International Carriage of Dangerous Goods by Inland Waterways; ADR - Agreement concerning the International Carriage of Dangerous Goods by Road; AIIIC - Australian Inventory of Industrial Chemicals; ASTM - American Society for the Testing of Materials; bw - Body weight; CLP - Classification Labelling Packaging Regulation; Regulation (EC) No 1272/2008; CMR - Carcinogen, Mutagen or Reproductive Toxicant; DIN - Standard of the German Institute for Standardisation; DSL - Domestic Substances List (Canada); ECHA - European Chemicals Agency; EC-Number - European Community number; ECx - Concentration associated with x% response; ELx - Loading rate associated with x% response; EmS - Emergency Schedule; ENCS - Existing and New Chemical Substances (Japan); ErCx - Concentration associated with x% growth rate response; GHS - Globally Harmonized System; GLP - Good Laboratory Practice; IARC - International Agency for Research on Cancer; IATA - International Air Transport Association; IBC - International Code for the Construction and Equipment of Ships carrying Dangerous Chemicals in Bulk; IC50 - Half maximal inhibitory concentration; ICAO - International Civil Aviation Organization; IECSC - Inventory of Existing Chemical Substances in China; IMDG - International Maritime Dangerous Goods; IMO - International Maritime Organization; ISHL - Industrial Safety and Health Law (Japan); ISO - International Organisation for Standardization; KECI - Korea Existing Chemicals Inventory; LC50 - Lethal Concentration to 50 % of a test population; LD50 - Lethal Dose to 50% of a test population (Median Lethal Dose); MARPOL - International Convention for the Prevention of Pollution from Ships; n.o.s. - Not Otherwise Specified; NO(A)EC - No Observed (Adverse) Effect Concentration; NO(A)EL - No Observed (Adverse) Effect Level; NOELR - No Observable Effect Loading Rate; NZIoC - New Zealand Inventory of Chemicals; OECD - Organization for Economic Co-operation and Development; OPPTS - Office of Chemical Safety and Pollution Prevention; PBT - Persistent, Bioaccumulative and Toxic substance; PICCS - Philippines Inventory of Chemicals and Chemical Substances; (Q)SAR - (Quantitative) Structure Activity Relationship; REACH - Regulation (EC) No 1907/2006 of the European Parliament and of the Council concerning the Registration, Evaluation, Authorisation and Restriction of Chemicals; RID - Regulations concerning the International Carriage of Dangerous Goods by Rail; SADT - Self-Accelerating Decomposition Temperature; SDS - Safety Data Sheet; SVHC - Substance of Very High Concern; TCSI - Taiwan Chemical Substance Inventory; TECI - Thailand Existing Chemicals Inventory; TRGS - Technical Rule for Hazardous Substances; TSCA - Toxic Substances Control Act (United States); UN - United Nations; vPvB - Very Persistent and Very Bioaccumulative

Further information

Other information : Observe national and local legal requirements
Items where changes have been made to the previous version
are highlighted in the body of this document by two vertical
lines.

Classification of the mixture:

Flam. Liq. 3	H226
Skin Irrit. 2	H315
Eye Dam. 1	H318
Aquatic Chronic 3	H412

Classification procedure:

On basis of test data.
Calculation method
Calculation method
Calculation method

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